

Robustness Reproducibility in Economics (R²E) - Pre-registration of robustness checks-

Name of replicator: Ryan McWay

Title of paper: Outsourcing Education: Experimental Evidence from Liberia

Original authors: Mauricio Romero, Justin Sandefur, Wayne Aaron Sandholtz

Below are the lists of potential analytical paths that could result from alternative choices for several dimensions of analytical decisions. The two tables below provide the potential challenges for the preferred specification presented by the authors of Romero et al. (2020). The two main outcomes of interest in this study were students' English test scores and mathematics test scores.

English Test Score:

Nr.	Main outcome from Table 3 Column 5	Analytical Decision	Rationale & Challenge	Choice(s) in Robustness Checks (include also the original analysis choice if that is considered one of the reasonable analytical choices)
1.	(1)	Sample	Given the there are several providers the implement treatment and education interventions often vary by gender.	1. Original 2. Sub-analysis by provider (Cannot be done due to restricted data) 3. Sub-analysis by gender
2.	(1)	Treatment	Some of the providers did not implement treatment for all schools	1. Original: Intention to Treat (ITT) 2. Treatment on the Treated (TOT)
3.	(1)	Controls	While the controls are prespecified, they may not be necessary and only a certain subset may actually change the results.	1. Original: prespecified controls 2. No controls 3. Only individual controls 4. Only school controls

4	(1)	Missing Observations	While the authors replace missing control values with a 0 and include a missing dummy, it may be that such a correction is not necessary.	1. Original: impute missing values 2. Do not replace missing values
5	(1)	Model	They have baseline data and could run an ANCOVA design (as mentioned in their PAP)	1. Original: RCT 2. ANCOVA
6	(1)	Outcome	There are several ways that student learning might be defined.	1. Original: IRT standardized scores 2. Initial/immediate student learning 3. Standardized scores by grade level 4. A PCA for student learning 5. Percentage of correctly answered questions.

Note: See Table 3 in the Reproducibility Assessment Protocol for an example.

Mathematics Test Score:

Nr.	Main outcome from Table 3 Column 5	Analytical Decision	Rationale & Challenge	Choice(s) in Robustness Checks (include also the original analysis choice if that is considered one of the reasonable analytical choices)
1.	(2)	Sample	Given the there are several providers the implement treatment and education interventions often vary by gender.	4. Original 5. Sub-analysis by provider (Cannot be done due to restricted data) 6. Sub-analysis by gender
2.	(2)	Treatment	Some of the providers did not implement treatment for all schools	3. Original: Intention to Treat (ITT) 4. Treatment on the Treated (TOT)
3.	(2)	Controls	While the controls are prespecified, they may not be necessary and only a certain subset may actually change the results.	5. Original: prespecified controls 6. No controls 7. Only individual controls 8. Only school controls
4	(2)	Missing Observations	While the authors replace missing control values with a 0 and include a missing dummy, it may be that such a correction is not necessary.	3. Original: impute missing values 4. Do not replace missing values
5	(2)	Model	They have baseline data and could run an ANCOVA design (as mentioned in their PAP)	3. Original: RCT 4. ANCOVA
6	(2)	Outcome	There are several ways that student learning might be defined.	6. Original: IRT standardized scores 7. Initial/immediate student learning

				8. Standardized scores by grade level 9. A PCA for student learning 10. Percentage of correctly answered questions.
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Note: See Table 3 in the Reproducibility Assessment Protocol for an example.